

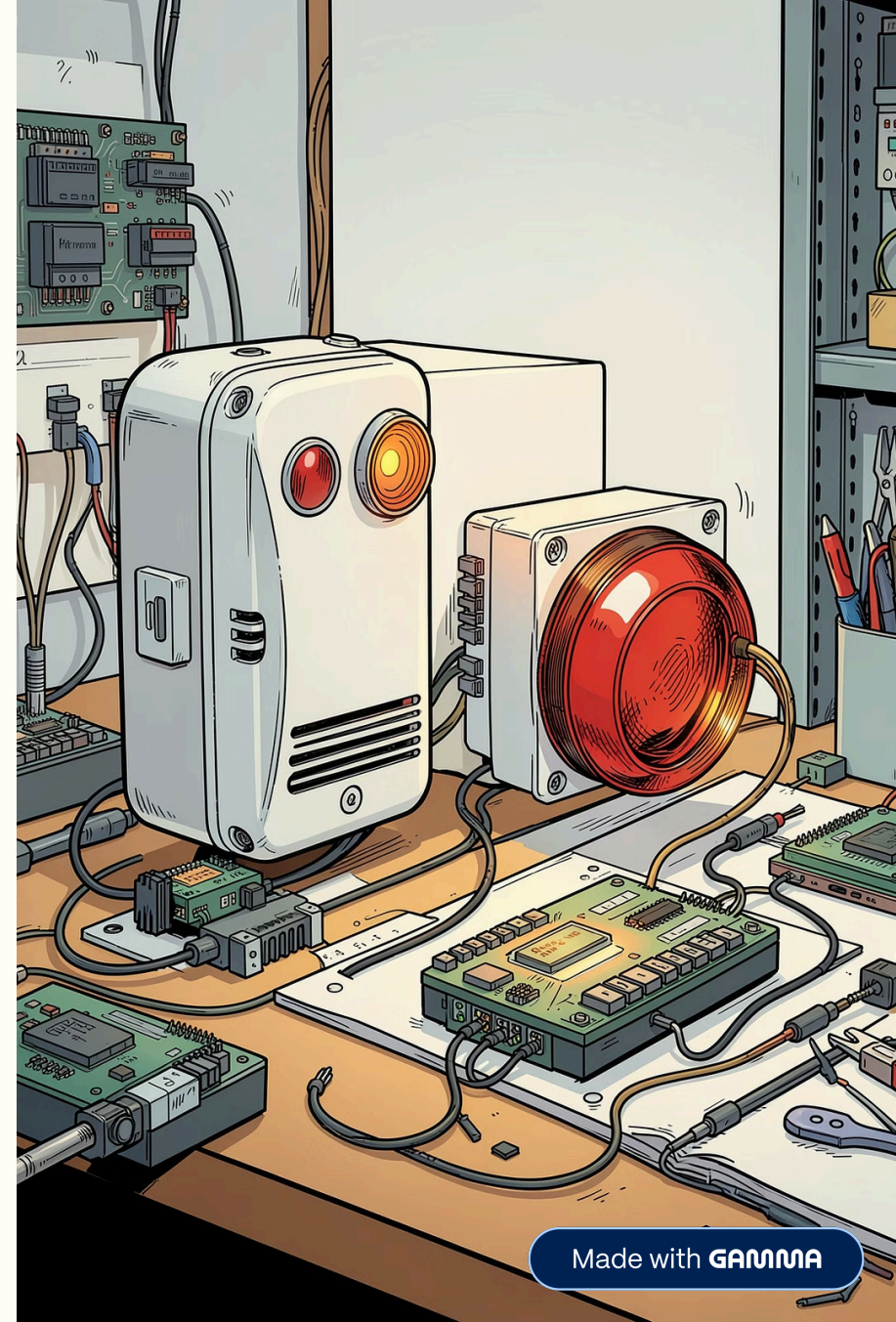
Fire Alarm Using a Flame Sensor

FIRST YEAR OF ENGINEERING (FE)

Basic Electronics Engineering project —

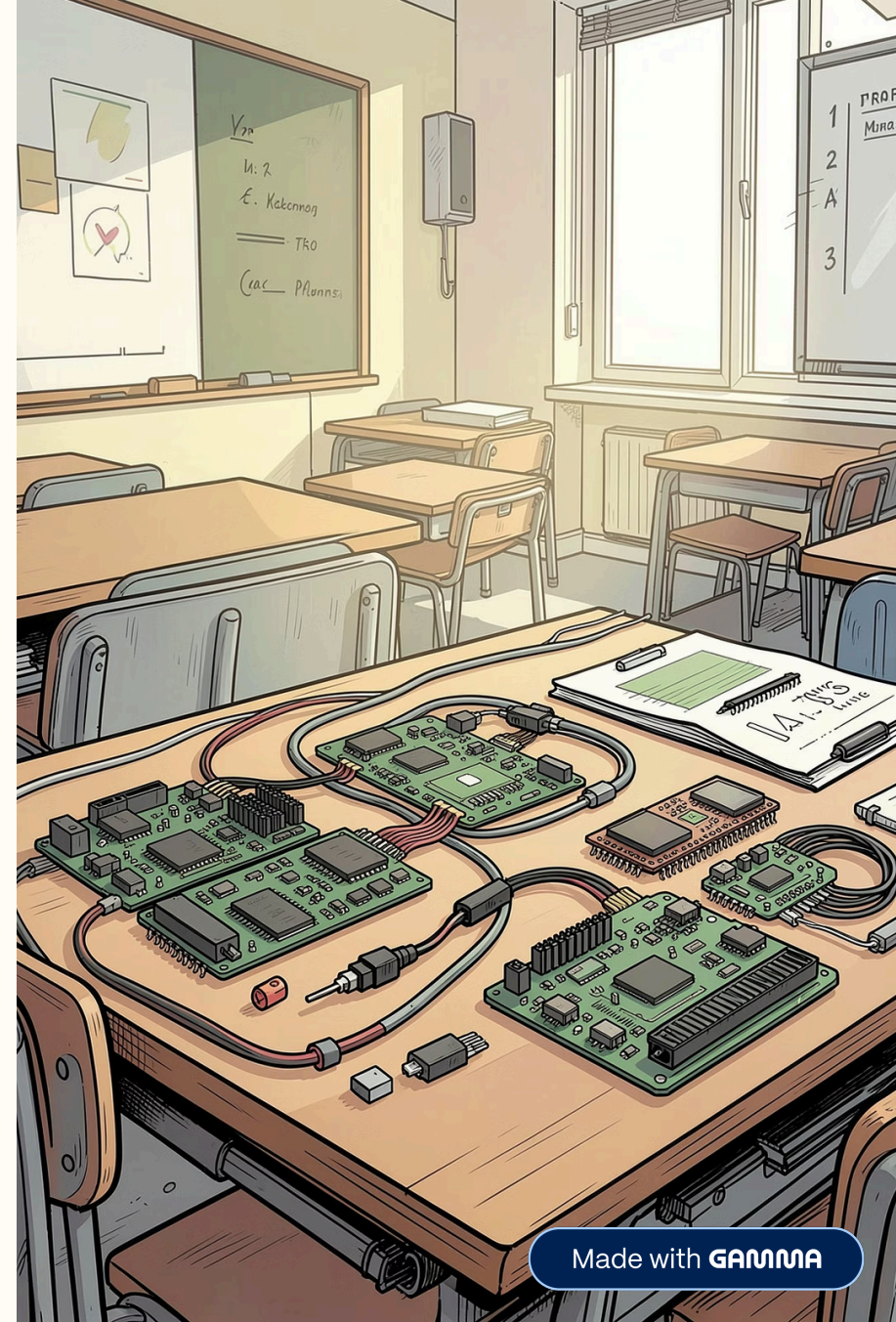
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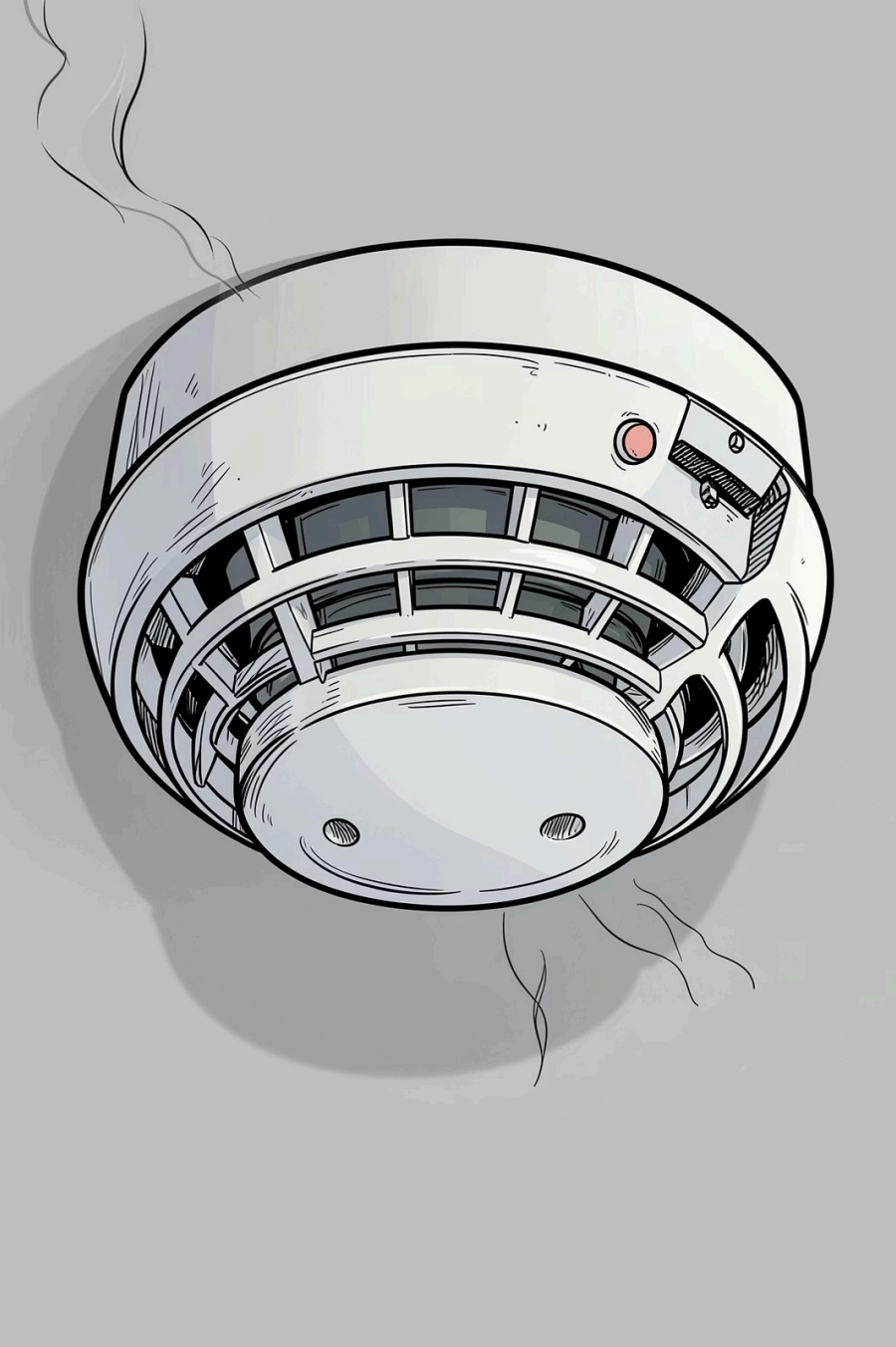
Subject teacher: Aasavari Bhalerao.



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Introduction

Fire alarms detect fires by sensing particles or flame signatures and trigger alerts so occupants can evacuate and call for help. Early detection prevents loss of life and property.

Overview

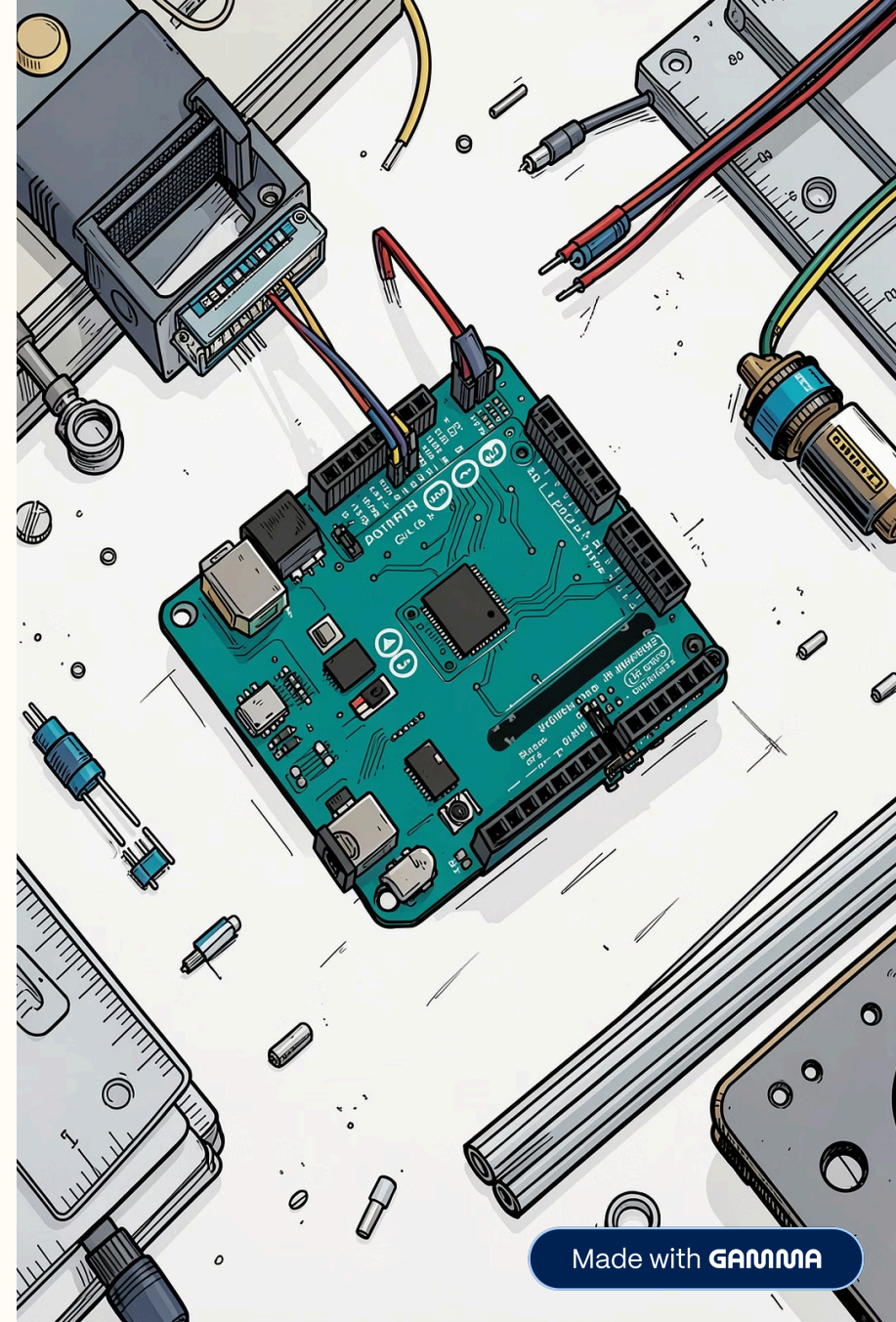
This project builds a fire detection system using a flame sensor and an Arduino (UNO R3/R4). The system detects flame presence and responds with a visual LED and a melody buzzer for immediate alerting.

Goal

Early, reliable flame detection with audible and visual alarms.

Approach

Use a YG1006-based flame sensor module with Arduino for detection and signaling.



Components Used



Arduino UNO

Microcontroller to read sensor and drive outputs.



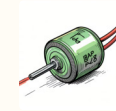
Flame Sensor (YQ1006)

Detects flame wavelengths (~760–1100 nm) via phototransistor.



Red LED

Visual indicator when flame is detected.



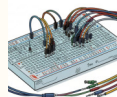
Buzzer

Audible alert (melody) driven by Arduino digital pin.



Type-C USB Cable

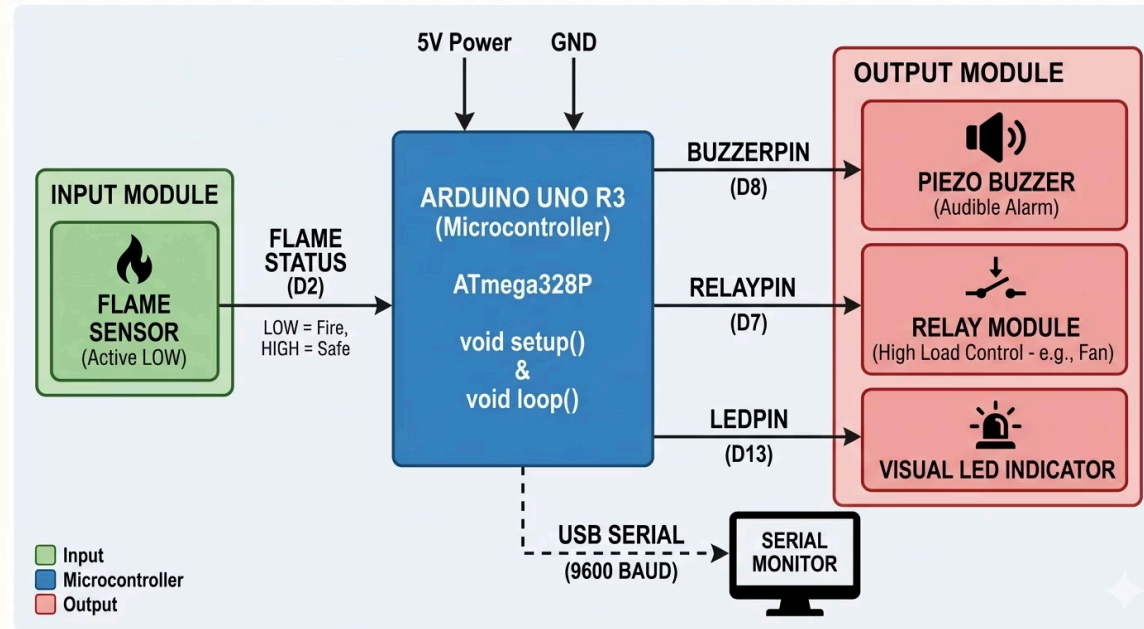
Power and programming connection for Arduino.



Breadboard & Wires

Prototype connections without soldering.

Block Diagram

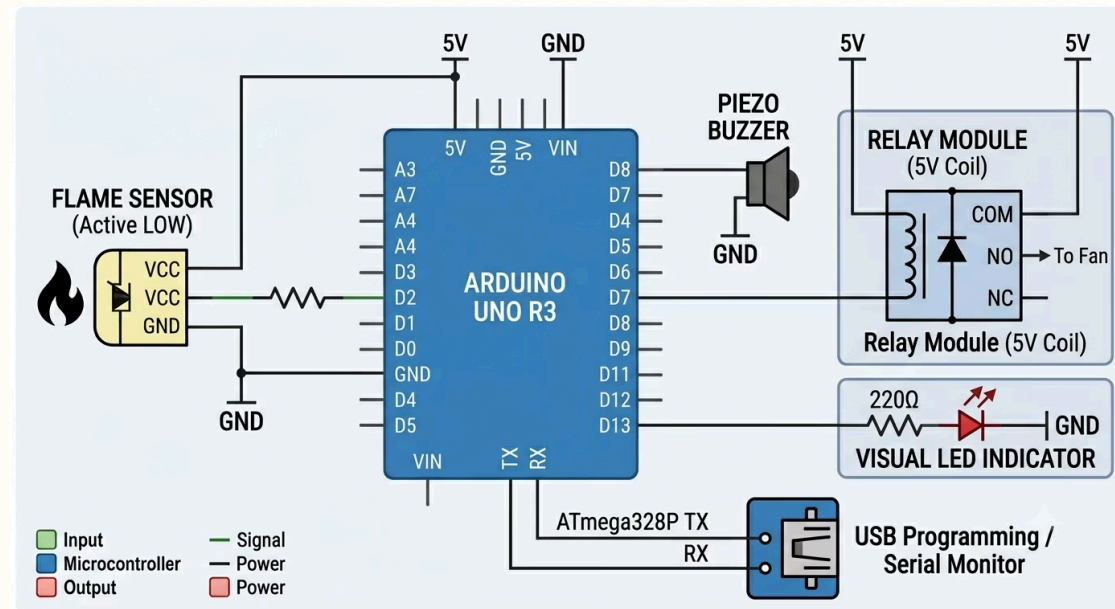


The system starts with the flame sensor, which detects infrared light from a fire and sends a signal to the Arduino.

The Arduino UNO R3 acts as the microcontroller. It reads the sensor input, decides whether a flame is present, and controls the output devices.

When fire is detected, the buzzer, relay module, and LED provide an audible and visual alert. The USB connection also sends status updates to the Serial Monitor for testing and debugging.

Circuit Diagram



The circuit is powered by the Arduino UNO, which supplies **5V** to the **VCC** pins of the connected components and provides a common **GND** connection for the entire system. Sharing the same ground is important because it gives all parts of the circuit a common reference point and allows signals to be read correctly.

The **flame sensor** is connected so that its **output pin goes to Arduino pin 4**. When the sensor detects infrared light from a flame, it sends a signal to the Arduino, which can then decide whether fire is present.

The **LED** is connected to **Arduino pin 13** through an appropriate resistor. The resistor is necessary to limit current and protect the LED from damage while still allowing it to light up as a visual indicator.

The **buzzer** is connected to **Arduino pin 12**. When the Arduino receives a flame detection signal, it can activate the buzzer to produce an audible alarm and warn the user.

If a **relay module** is used, it is also connected to the Arduino so that the system can control an external load such as a lamp, fan, or pump. The relay acts like an electronically controlled switch, allowing the Arduino to safely turn another device on or off.

Together, these connections let the Arduino monitor the flame sensor and respond with both visual and sound alerts. If a fire is detected, the Arduino can activate the LED, buzzer, and relay module, creating a simple but effective student fire detection system.

Flame Sensor & How It Works



Module Anatomy

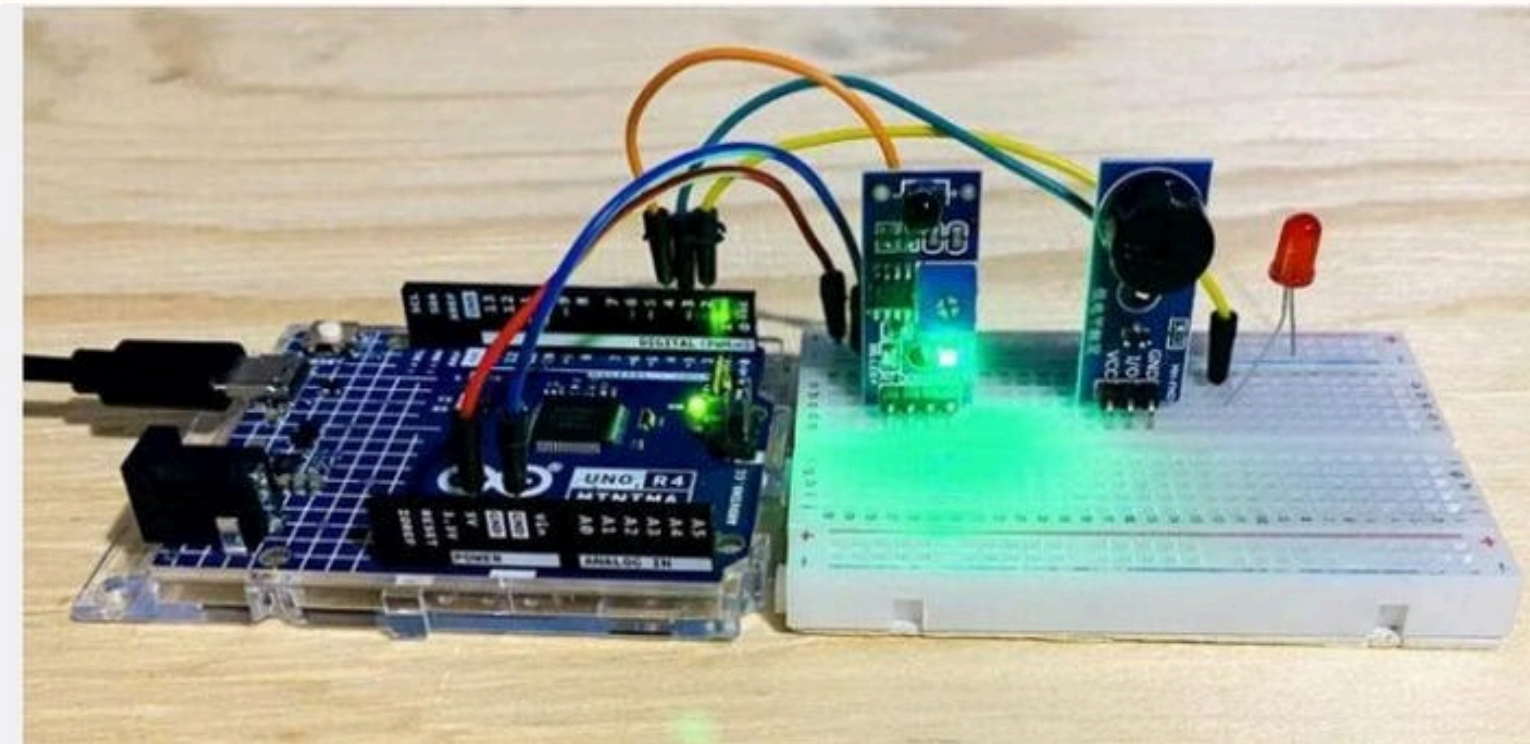
- Photodiode / phototransistor (detects IR)
- Potentiometer (10K) for sensitivity
- LM393 comparator, indicator LED, output pin (DO)
- VCC (3–5.5V) and GND

IR photons change detector response; comparator outputs a digital signal indicating flame presence.

Testing & Results

Behavior observed during testing:

- Flame detected → sensor outputs DIGITAL LOW to Arduino pin 4.
- Arduino responds: sets LED pin HIGH (visual) and activates buzzer melody on buzzer pin.
- No flame → sensor outputs HIGH → Arduino turns LED off and stops buzzer.



Applications, Conclusion & Acknowledgement

Applications

Homes, offices, schools, labs, furnaces—standalone or integrated into security systems for early warning.

Conclusion

A simple Arduino + flame sensor system provides reliable early detection and immediate alerts, demonstrating how basic components can increase safety.

Acknowledgement

Special thanks to our teacher, Aasavari Bhalerao, for guidance and support.

